

The effects of three cardioplegia solutions on myocardial metabolism during ischaemia-reperfusion in a Langendorff murine model

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Objectives: To compare the impact of ischaemia and ischaemia-reperfusion on myocardial metabolism following cardioplegic arrest using three cardioplegia solutions in common clinical use during cardiac surgery.

Methods: Isolated Langendorff-perfused mouse hearts were exposed to 40 minutes stabilisation +/- 45 minutes global ischaemia with or without cardioplegia +/- 60 minutes reperfusion (Figures 1 & 2). One of three types of cardioplegia was perfused at the onset of ischaemia: del Nido (DNB), Custodiol (HTK) or St. Thomas' Hospital No.2 blood (STB). Hearts were snap-frozen and stored prior to tissue homogenisation and metabolite extraction, applying a modified Bligh-Dyer biphasic solvent mixture to generate two solutions: a water/methanol solution containing water-soluble metabolites and a chloroform solution containing lipids. All samples were analysed applying two complementary untargeted ultra-performance liquid chromatography-mass spectrometry assays. Raw data was processed applying XCMS and PUTMEDID followed by univariate and multivariate analysis in MetaboAnalyst.

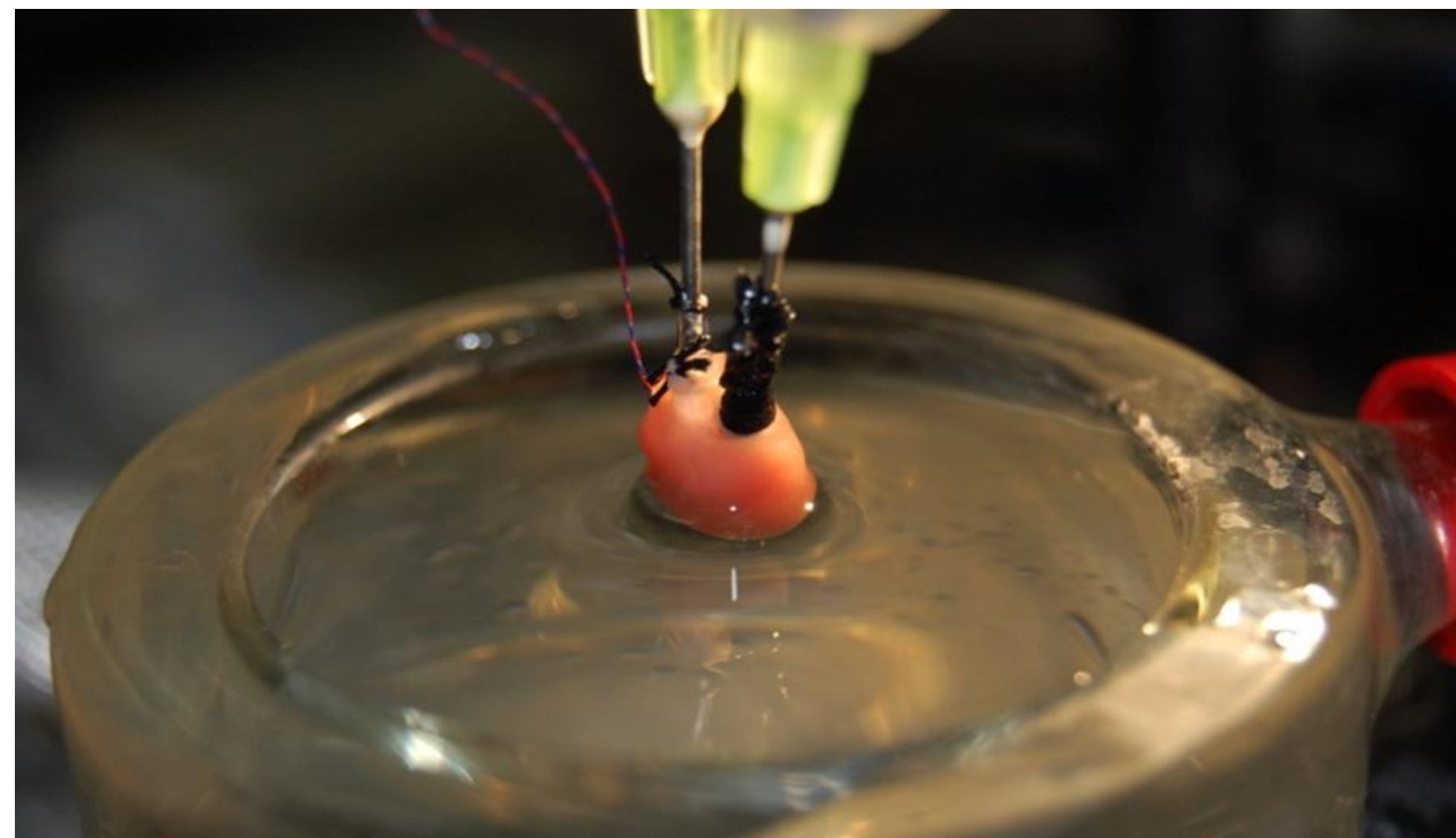


Figure 1. Langendorff model of ischaemia-reperfusion in isolated mouse hearts, including aortic perfusion cannula, pacing wire and water bath.

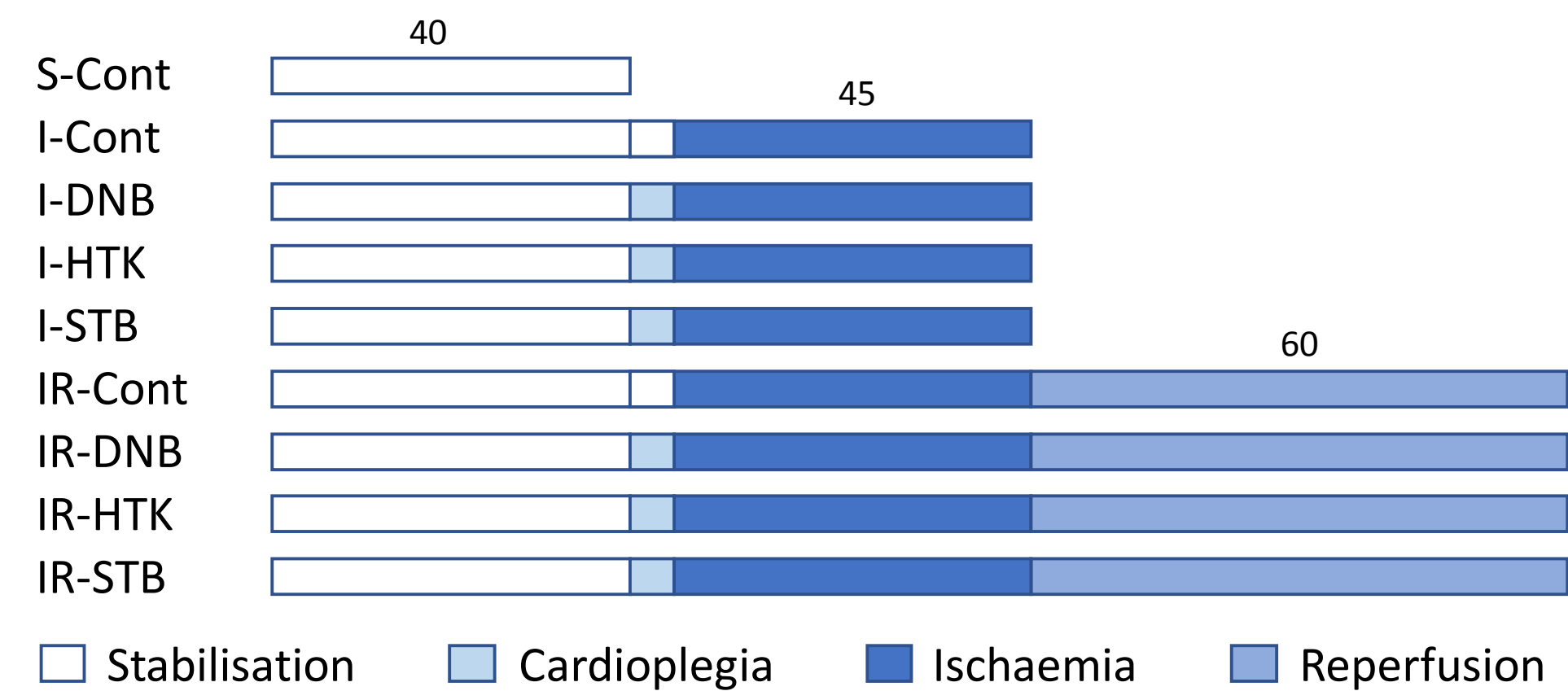


Figure 2. Experimental protocol for ischaemia-reperfusion (n=5 per group)

Results: Differences in the concentration of metabolites between ischaemia and reperfusion (independent of cardioplegia type) and between the three different cardioplegia solutions at the end of ischaemia (figure 3) and after reperfusion (figure 4) were observed. The major metabolic impact was observed at the end of ischaemia (626 metabolites with $q < 0.05$, table 1) with a smaller number of metabolic differences following reperfusion (127 metabolites with $q < 0.05$). Significant changes were observed in glucose oxidative metabolism (glycolysis, TCA cycle, pantothenate and CoA metabolism), proteolysis and peptide signalling, fatty acid oxidation (acyl carnitines, fatty acids, oxidised fatty acids) and purines & pyrimidine metabolism.

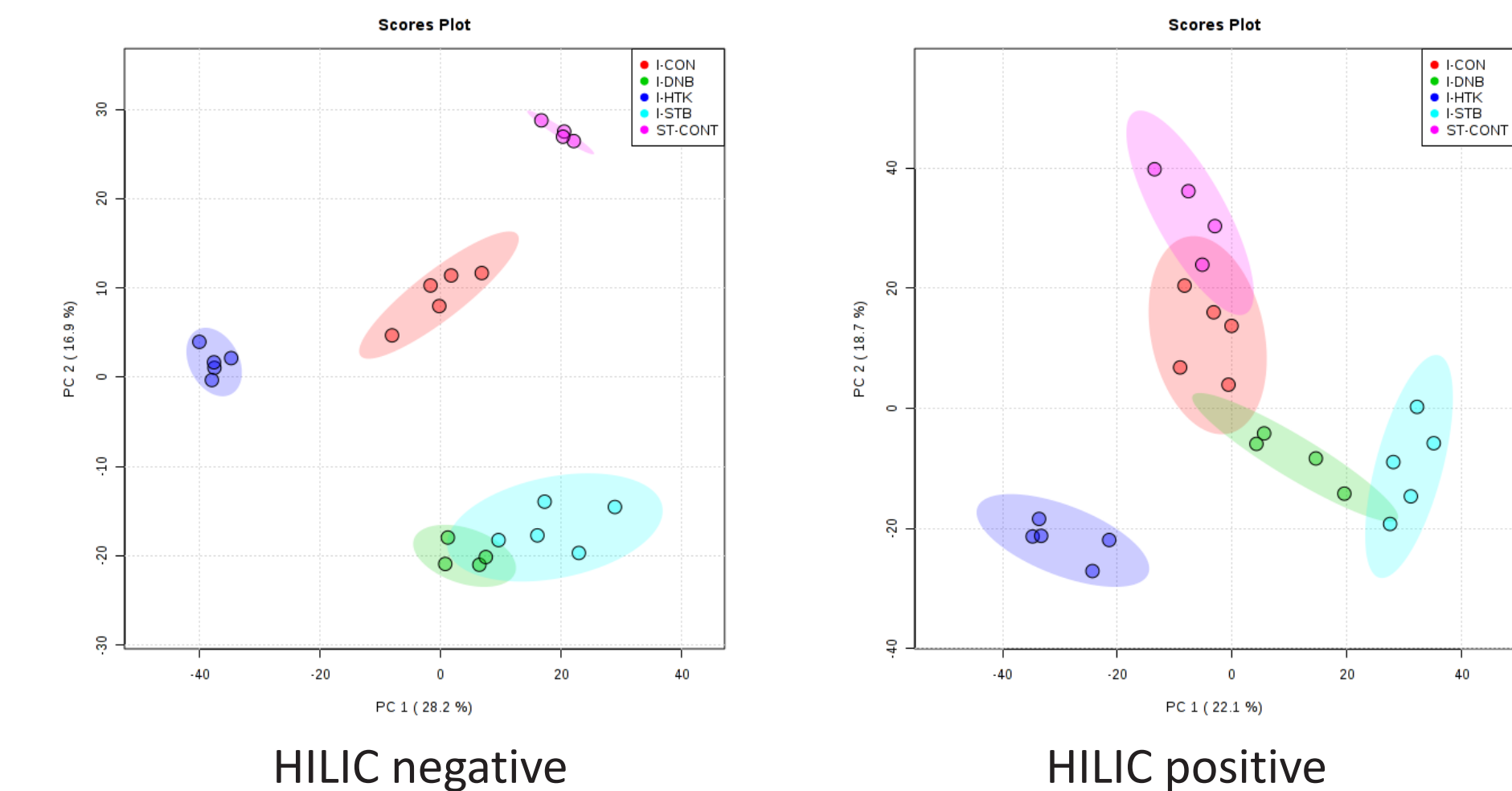


Figure 3. PCAs for water-soluble metabolites at the end of ischaemia.

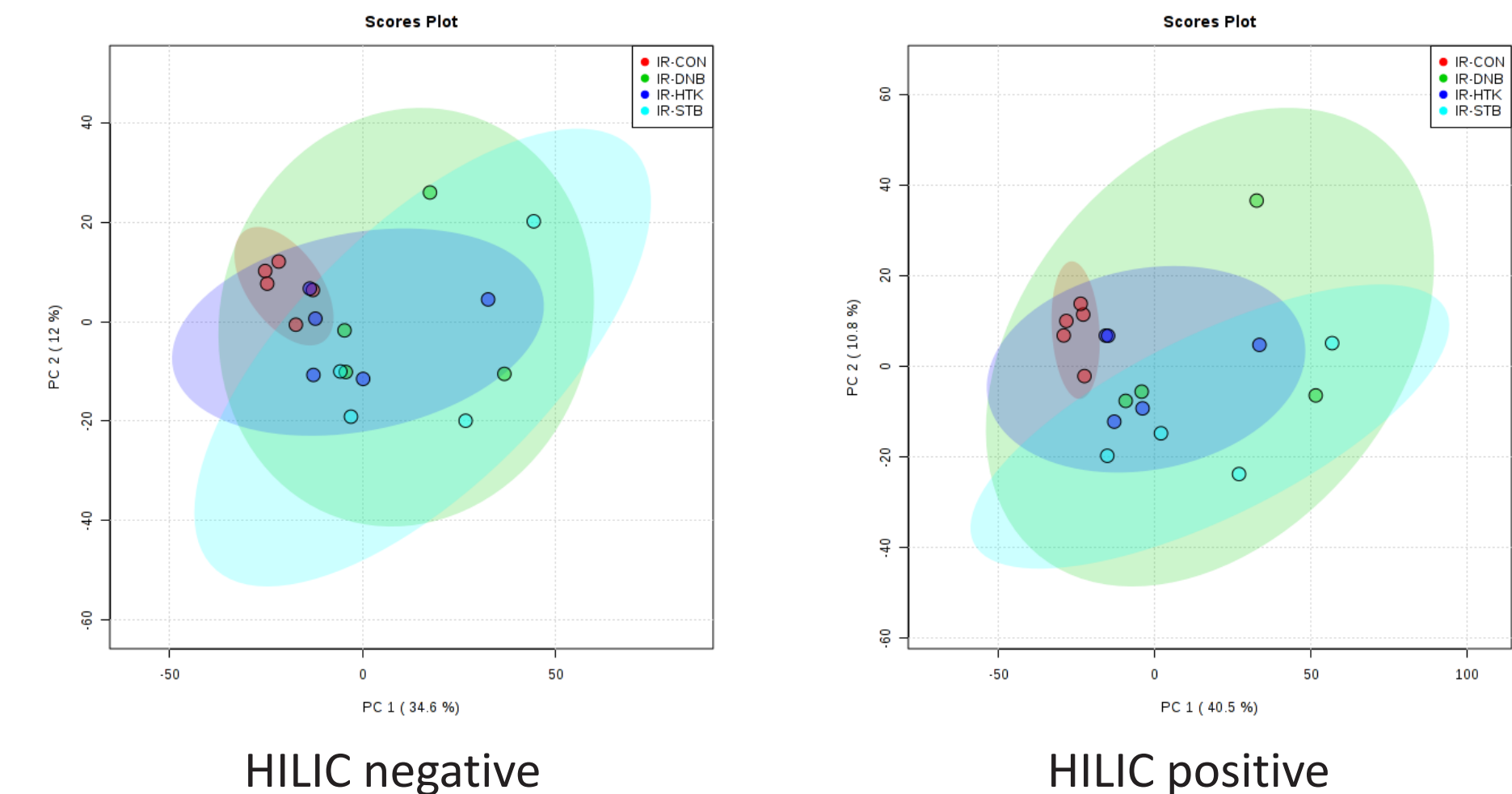


Figure 4. PCAs for water-soluble metabolites following reperfusion.

Metabolic pathway	No. metabolites
Peptide	67
Acyl carnitine	50
Purine & pyrimidine	41
Sulphur metabolism	40
Lysoglycerophospholipids	35
Carboyhdrates	17
Fatty acids	17
Tryptophan metabolism	15
Oxidised fatty acids	13
Acyl amino acids	10
Creatine metabolism	3

Table 1. Univariate analysis of differences between metabolites at the end of ischaemia with the three cardioplegia solutions

Conclusions: The type of cardioplegia solution used for myocardial protection has a significant effect on tissue metabolism, with a greater impact at the end of ischaemia than following reperfusion. Future studies are warranted to link metabolic changes in the myocardium to clinical outcomes.